

How Much of Deep Joint Source–Channel Coding’s Advantage Survives a Fair Comparison? Evidence from Image Classification

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Abstract—Deep joint source–channel coding (DJSCC) offers a different way to transmit images over bandwidth-constrained wireless channels. Instead of separately compressing an image and protecting the resulting bits against channel errors, DJSCC learns the source representation and its transmission jointly. For task-oriented applications, where the receiver may only need to classify an image rather than reconstruct it perfectly, this raises an important question: does the benefit come from joint source–channel coding itself, or simply from learning to transmit information that is more useful for the task?

This paper develops a controlled three-system framework for studying that question over a complex additive white Gaussian noise channel. The conventional digital system transmits JPEG 2000-compressed images using an NR-LDPC-based physical layer. A second, task-aware digital control uses the same transmission chain but replaces the conventional image representation with quantized learned features. The third system is an end-to-end DJSCC model that maps images directly to channel symbols and supports both reconstruction and classification at the receiver.

The three systems are given the same transmission budget and are tested under the same channel conditions and dataset splits. This lets us compare them more fairly and see whether any improvement comes from learning better features for classification, from learning how to transmit those features over the channel, or from both.

We report validation results on 1000 Imagenette-160 images at 21 SNRs from -8 to 18 dB. At a bandwidth ratio of $1/6$, DJSCC classifies 72.8% of images correctly at -8 dB, where neither digital system delivers any packets. Between -5 and -4 dB both digital systems start delivering, and from -4 dB upward the JPEG 2000 system is more accurate than DJSCC, reaching 89.3% against 83.4% at 18 dB. The task-aware digital system reaches 82.0% once its packets arrive. At a ratio of $1/24$ the JPEG 2000 system first overtakes DJSCC at $+4$ dB, compared with -4 dB at $1/6$. The receiver’s classifier matters as much as the link: scoring the same JPEG 2000 outputs with a classifier trained only on uncompressed images costs up to 43 points near the delivery threshold and widens the SNR range in which DJSCC leads by 3 dB at $1/6$ and by 6 dB at $1/24$. The held-out test split has not been used; those results will come from a single preregistered evaluation over three trained seeds.

Index Terms—semantic communication, task-oriented communication, deep joint source–channel coding, image classification, 5G NR LDPC, JPEG 2000

I. INTRODUCTION

Conventional communication systems usually treat source coding and channel coding as two separate problems. The source is first compressed into a digital bitstream. Redundancy is then added so that the bitstream can be protected against

errors introduced by the channel. Shannon’s separation theorem provides the classical foundation for this design: under its assumptions and with sufficiently long blocklengths, source and channel coding can be optimized independently without losing optimality [1], [2].

Real systems do not operate in this asymptotic setting. They use finite blocklengths and must meet finite latency constraints. As a result, coding-rate back-off and nonzero decoding-failure probabilities matter in practice [3], [4]. Packetization and format overhead also consume part of the available transmission budget, leaving less room for the source payload itself.

This distinction becomes especially important when the receiver does not need a bit-perfect reconstruction of the original source. In machine-oriented communication, the real goal may be a downstream task such as classification, retrieval, or control. In that setting, transmitting every visual detail of an image may be unnecessary. A lower-dimensional representation can be sufficient if it preserves the information needed for the task. This motivates *task-oriented*, or more broadly semantic, communication systems, where the transmitted representation is optimized for the receiver’s objective rather than only for reconstruction quality.

Deep joint source–channel coding is one way to build such a system. Instead of first producing a compressed bitstream and then protecting it with a separate channel code, DJSCC uses a neural encoder to map the source directly to channel symbols. A differentiable channel model is placed between the transmitter and receiver during training, allowing the neural encoder and decoder to be optimized end to end [5]. Early work on image transmission showed the well-known graceful-degradation behavior of neural JSCC [5]. Later studies extended the idea to feedback, variable bandwidth, OFDM-oriented models, and transformer-based architectures [6], [7], [8], [9].

A simple learned-versus-digital comparison does not reveal why the learned system performs differently. A task-oriented DJSCC encoder changes two things at the same time. First, it can learn a representation that is tailored to the task. Second, it removes the usual boundary between source coding and channel coding. A conventional JPEG-based system changes neither of these: it sends a generic image representation and relies on a separate channel code for robustness.

This creates an attribution problem. If the learned system achieves higher task accuracy, the gain may come from the

learned representation, from joint source–channel optimization, from limitations of the chosen digital baseline, or from some combination of these factors. For this reason, a controlled comparison needs a task-aware digital system in addition to the conventional reconstruction-oriented baseline.

There is also a second problem: the digital baseline must be given the same physical transmission budget as the learned system. Calling a baseline simply “JPEG + LDPC” is not enough, because finite-block overhead and the selected coding and modulation settings determine how much useful source data can actually be carried within a fixed number of complex channel uses. A digital baseline can also be unfairly weakened if it is forced to use a single operating point across every SNR.

To address both issues, the study is structured around three systems for task-oriented image classification. The first is a reconstruction-oriented separated baseline using JPEG 2000, a TS 38.212-derived 5G NR LDPC coding/rate-matching core, and BPSK, QPSK, or 16-QAM [10], [11]. The second is a task-oriented separated digital control in which learned image features are quantized and sent through the same digital physical layer. The third is a task-aware DJSCC system that maps images directly to a fixed number of complex channel symbols and produces both reconstruction and classification outputs at the receiver. All three systems are implemented and evaluated on the validation split in this manuscript. The held-out test evaluation has not yet been run.

The study therefore asks how much of DJSCC’s advantage for classification survives when the digital baseline is given a fair chance. To our knowledge, it makes three contributions:

- a three-system comparison in which a reconstruction-oriented digital link and a task-aware digital link share one finite-block NR-LDPC chain, and the learned JSCC system uses the same complex-channel-use budget and E_s/N_0 convention;
- a measurement of how strongly the receiver’s classifier sets the outcome: on validation data, scoring the same digital outputs with a classifier trained on uncompressed images instead of one fine-tuned on compressed images changes accuracy near the digital threshold by up to 43 points and widens the SNR range in which DJSCC leads by 3 to 6 dB;
- an attribution of DJSCC’s low-SNR advantage: at the main bandwidth ratio, learned task features sent over the same digital link come within about two points of DJSCC across three seeds wherever their packets arrive, so most of the advantage there comes from avoiding outage rather than from a better task representation.

None of the three systems is new on its own. The contribution lies in the controls and in what they show about the comparison.

The experiments in this manuscript use image classification over a simulated complex AWGN channel. Top-1 classification accuracy is the primary task metric, while reconstruction quality is retained as a secondary diagnostic.

II. RELATED WORK

A. Deep JSCC and practical digital comparators

DeepJSCC introduced end-to-end image transmission in which a neural encoder maps images directly to noisy-channel symbols instead of first producing a reliable intermediate bitstream [5]. Later work extended the approach to feedback and bandwidth adaptation [6], [7]. Constellation-constrained variants further showed that learned JSCC can operate with finite channel-input alphabets [12]. Practical digital comparators have also appeared in earlier studies, where conventional image codecs were combined with finite-length LDPC coding and digital modulation rather than compared only through idealized channel-capacity limits [6], [12]. Transformer-based JSCC has since been compared with BPG and 5G NR LDPC coding, with the modulation and code rate chosen separately at each SNR [13], and other work argues that well-engineered separate source and channel coding remains competitive with JSCC [14]. The present work therefore does not claim novelty for using a practical channel code or for comparing digital transmission with DeepJSCC.

B. Task-oriented and digital semantic communication

Task-oriented communication changes the receiver’s objective from faithful reconstruction to a downstream inference task. For example, Jankowski *et al.* compared learned digital feature transmission with direct learned feature-to-channel mappings for wireless image retrieval [15]; their digital scheme assumed capacity-achieving channel codes. Information-bottleneck methods have been used to learn compact task-oriented features for edge inference [16], including a robust variant that encodes features into a discrete representation for digital modulation [17]. Huang *et al.* sent entropy-coded learned image features over a practical LDPC code, alongside BPG over the same code and DJSCC, for joint reconstruction and classification; the learned digital scheme led DJSCC wherever its bits decoded reliably and showed the digital cliff below that [18]. Later work adapts the modulation order of such digital systems to the channel [19] and improves their vector quantizer [20]. A task-aware digital arm is therefore not a new communication paradigm. In this study, its role is instead to act as an attribution control while sharing the same implemented digital physical layer as the reconstruction-oriented arm.

Using a learned JSCC receiver for both reconstruction and classification is also established in prior work. Lyu *et al.* study semantic communication for simultaneous image recovery and classification [21], while later work examines how reconstruction-oriented and task-oriented communication objectives can be aligned [22]. Recent vector-quantized semantic systems have also compared conventional image coding, learned digital representations, and learned JSCC with practical channel coding [23]. Accordingly, this paper does not claim the first three-family comparison or a novel dual-head architecture.

C. Positioning of this study

The contribution pursued here is mainly methodological. Unlike a simple learned-versus-digital comparison, the study con-

TABLE I
IMAGENETTE-160 SPLIT SIZES USED IN THE STUDY.

Dataset	Train	Val.	Test
Imagenette-160	8469	1000	3925

trols both the transmitted representation and the choice between separated and joint source–channel coding. A reconstruction-oriented digital link and a task-oriented digital feature link share the same finite-block NR-LDPC/rate-matching chain, while the learned JSCC arm uses the same complex-channel-use budget and E_s/N_0 convention. This arrangement is intended to make differences between the three systems easier to attribute to the representation and coding strategy rather than to unequal transmission resources.

The study also controls the classifier at the receiver. Recognition models lose accuracy on compressed images mainly because they have not seen compression artifacts, and retraining them on compressed images recovers most of that loss [24]. Learned-versus-digital comparisons nevertheless may classify the decoded digital images with a network trained on uncompressed images, as in [18]. Here the digital outputs are scored by both a clean and an artifact-fine-tuned classifier, so the effect of that choice on the comparison can be measured.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Source and channel-use budget

Consider an image-label pair $(\mathbf{x}, y) \sim \mathcal{D}$, where $\mathbf{x} \in [0, 1]^{H \times W \times C}$ is an RGB image and $y \in \{1, \dots, M\}$ is its class label. For Imagenette-160 [25], $H = W = 160$, $C = 3$, and $M = 10$. The source dimensionality is $n = HWC$. A communication system is allocated k complex baseband channel uses per image, with bandwidth ratio

$$r = \frac{k}{n}. \quad (1)$$

The configured bandwidth-ratio set is

$$r \in \left\{ \frac{1}{2}, \frac{1}{3}, \frac{1}{6}, \frac{1}{12}, \frac{1}{24}, \frac{1}{48} \right\}. \quad (2)$$

For Imagenette-160, $n = 76,800$, giving channel-use budgets of 38,400, 25,600, 12,800, 6,400, 3,200, and 1,600, respectively.

B. Channel model

Let $\mathbf{z} \in \mathbb{C}^k$ denote the transmitter output before power normalization. The channel input is normalized independently per image as

$$\tilde{\mathbf{z}} = \frac{\mathbf{z}}{\sqrt{\frac{1}{k} \sum_{i=1}^k |z_i|^2}}, \quad (3)$$

which imposes unit average symbol power. Transmission uses a memoryless complex AWGN channel,

$$\mathbf{r} = \tilde{\mathbf{z}} + \gamma^{-1/2} \mathbf{n}, \quad \mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}), \quad (4)$$

where $\gamma = 10^{\text{SNR}_{\text{dB}}/10}$. SNR is reported as E_s/N_0 per normalized complex channel use. For digital references stated in E_b/N_0 ,

$$\left. \frac{E_s}{N_0} \right|_{\text{dB}} = \left. \frac{E_b}{N_0} \right|_{\text{dB}} + 10 \log_{10}(Q_m R_c), \quad (5)$$

where Q_m is coded bits per modulation symbol and R_c is channel-code rate.

To make paired comparisons more controlled, the implementation uses a system-independent keyed noise identity. For a fixed image, SNR, seed, and channel-use budget, paired systems therefore receive the same deterministic AWGN realization whenever the representation permits such pairing.

C. Task objective

Each system ultimately produces class logits $\mathbf{s} \in \mathbb{R}^M$ and prediction $\hat{y} = \arg \max_j s_j$. The primary endpoint is top-1 task accuracy as a function of SNR,

$$A(\gamma) = \Pr(\hat{y} = y \mid \gamma). \quad (6)$$

PSNR and SSIM are retained as secondary reconstruction diagnostics, not substitutes for the downstream task endpoint.

IV. COMPARED COMMUNICATION SYSTEMS

A. Reconstruction-oriented separated digital system

The conventional arm is

$$\begin{aligned} \mathbf{x} &\xrightarrow{\mathcal{C}_{\text{J2K}}} \mathbf{b}_s \xrightarrow{\mathcal{P}} \mathbf{b}_p \xrightarrow{\mathcal{E}_{\text{LDPC}}} \mathbf{b}_c \xrightarrow{\mathcal{M}} \mathbf{z} \\ &\xrightarrow{\text{AWGN}} \mathbf{r} \xrightarrow{\mathcal{D}_{\text{digital}}} \hat{\mathbf{x}} \xrightarrow{h} \hat{y}. \end{aligned} \quad (7)$$

Here $\mathcal{C}_{\text{J2K}}$ denotes JPEG 2000 source coding, $\mathcal{P}$ is our transport-block accounting and packetization logic, $\mathcal{E}_{\text{LDPC}}$ is the TS 38.212-derived LDPC coding/rate-matching chain, and $\mathcal{M}$ represents BPSK, QPSK, or 16-QAM. Sionna 2.0.1 [26] provides the LDPC encoder and decoder. Segmentation, CRC, filler handling, and transport accounting remain explicitly represented by the experiment.

For a chosen modulation, the coded-bit budget is $G = kQ_m$. The source payload is then chosen as the largest byte-aligned transport-block payload whose full coding and rate-matching representation fits inside G . Failed transmissions are not removed from evaluation. Structural infeasibility, codec infeasibility, decode failure, and successful delivery are recorded as distinct outcomes, and the prescribed outage outcome remains in the task-accuracy denominator.

B. Task-aware learned JSCC system

The learned joint system uses an encoder

$$f_\theta : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{C}^k, \quad (8)$$

followed by the common power normalization and AWGN model. The encoder has two strided 5×5 convolutions with 64 and 128 channels, which downsample the image by a factor of four to a 40×40 grid, then two residual blocks with group normalization [27] and PReLU activations [28]. A final 3×3 convolution produces $2c$ real feature channels, and each

pair becomes one complex symbol. The value of c sets the bandwidth ratio: $c = 8$ gives $1600 \times 8 = 12,800$ symbols at $r = 1/6$, and $c = 2$ gives 3200 symbols at $r = 1/24$.

The receiver has a reconstruction head and a classification head,

$$g_\phi(\mathbf{r}) = (\hat{\mathbf{x}}, \mathbf{s}). \quad (9)$$

Both heads share a decoder trunk that mirrors the encoder. The reconstruction head upsamples with two transposed convolutions. The classification head applies global average pooling to the 128-channel trunk output, followed by one linear layer. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\mathbf{s}, y) + \lambda \mathcal{L}_{\text{MSE}}(\hat{\mathbf{x}}, \mathbf{x}). \quad (10)$$

The weight λ was chosen from $\{0, 0.1, 0.3, 1, 3\}$ in a validation pilot at 7 dB, using a preregistered rule with a one-point accuracy tolerance and a PSNR floor on the reconstruction. The rule selected $\lambda = 3$. The model has 1,567,197 parameters at $r = 1/6$ and 1,539,537 at $r = 1/24$.

C. Task-oriented separated digital feature control

The attribution control is

$$\mathbf{x} \xrightarrow{q_\psi} \mathbf{u} \xrightarrow{\text{quantize}} \mathbf{b}_u \xrightarrow{\text{same digital link}} \hat{\mathbf{b}}_u \xrightarrow{d_\omega} \hat{y}. \quad (11)$$

The feature encoder q_ψ reuses the residual trunk of the DJSCC encoder. Its output is projected to a D -dimensional vector, and each element is quantized by a uniform 2-bit scalar quantizer. The quantized indices are range-coded with a static entropy model fitted on the training set. If range coding would produce more bits than the raw $2D$ -bit representation, the raw form is sent instead, with a one-bit flag marking the choice. The message is then zero-padded to the transport-block payload and sent through the same CRC, LDPC, rate-matching, and modulation chain as the JPEG 2000 system, using the same k channel uses. At the receiver, the decoded indices are dequantized and passed to a task head d_ω .

The dimension D was selected on validation data from $\{64, 128, \dots, 2048\}$. Each candidate was trained and then evaluated through the real digital chain at 7 dB. Accuracy rose with D from 28.4% at $D = 64$ to 82.0% at $D = 2048$, which was selected. With $D = 2048$ the raw payload is 4096 bits. It fits a BPSK, rate-1/3 transport block at $r = 1/6$, and that physical-layer setting is used at every SNR.

This system keeps task-aware representation learning but puts back a discrete boundary between source and channel coding. The three-way design therefore separates reconstruction-oriented digital transmission, task-aware digital transmission, and task-aware joint transmission. Comparing the first two measures the effect of changing the transmitted representation toward the task. Comparing the task-aware digital system with DJSCC measures what joint source-channel optimization adds on top of that. The feature encoders, quantization, and task heads differ between the two learned systems, so this second comparison is not a pure channel-coding contrast.

TABLE II
SYSTEM VARIANTS EVALUATED ON THE VALIDATION SPLIT. EVERY VARIANT USES THE SAME COMPLEX AWGN CHANNEL AND THE SAME k FOR ITS BANDWIDTH RATIO.

Variant	r	Classifier at receiver
DJSCC (7 dB training)	1/6, 1/24	Own task head
DJSCC, SNR-randomized	1/6	Own task head
DJSCC, PAPR-capped	1/6	Own task head
DJSCC reconstruction	1/6	Clean ResNet-18
JPEG 2000 + LDPC, adaptive	1/6, 1/24	Fine-tuned ResNet-18
JPEG 2000 + LDPC, QPSK only	1/6	Fine-tuned ResNet-18
JPEG 2000 + LDPC, fixed MCS	1/6	Fine-tuned ResNet-18
Baseline JPEG + LDPC	1/6	Fine-tuned ResNet-18
Task-aware digital	1/6	Own task head
Label transmission	1/6	Transmitter's prediction

D. Additional variants

Table II lists every system variant evaluated. Besides the three main systems, the study includes:

- *SNR-randomized DJSCC*, with the same architecture, trained with an SNR drawn per sample and per epoch from $\{1, 4, 7, 13, 19\}$ dB;
- *PAPR-capped DJSCC*, with the same architecture plus a projection after power normalization that limits the symbol-domain PAPR to 3 dB. The projection preserves each symbol's phase and finds, by bisection, a common scaling that keeps unit mean power while respecting the peak bound;
- a *reconstruction ablation*, in which the DJSCC reconstruction $\hat{\mathbf{x}}$ is classified by the clean-trained ResNet-18 instead of the DJSCC task head;
- two restricted JPEG 2000 baselines: one fixed to QPSK with the rate and resolution still adapted, and one fixed to a single operating point (16-QAM, LDPC rate 1/2, full resolution) at every SNR;
- a *baseline JPEG* system, which replaces JPEG 2000 with baseline JPEG and is selected in the same way;
- a *label transmission* system, in which the transmitter runs the task-aware digital model to completion and sends only its predicted label, a 4-bit field in a one-byte frame, as the first byte of the same packet used by the task-aware digital system.

E. Receiver classifiers

Two ResNet-18 classifiers [29] read images at the receiver. The clean classifier is trained from scratch on the uncompressed Imagenette training images. The fine-tuned classifier starts from the clean one and is trained for a further 20 epochs on 44,039 JPEG 2000-decoded versions of the training images. The compression settings are drawn from the operating points the baseline can select at or below 7 dB. The fine-tuned classifier is the primary scorer for every JPEG 2000 and JPEG variant, because a classifier that has never seen compression artifacts would understate what the digital system delivers. Scores from the clean classifier on the same received images are also kept and are reported in Section VII-D.

V. EXPERIMENTAL PROTOCOL

A. Datasets and split discipline

Imagenette-160 is used for the experiments, with 8469 training, 1000 validation, and 3925 test images across 10 classes. Dataset membership and sample identities are fixed through versioned manifests. The published training partition is deterministically divided into project training and validation subsets, while the published Imagenette validation partition is used as the project test split.

Model and operating-point choices use only the training and validation data. The project test split is excluded from model selection and digital operating-point selection. It has not been read by any model or selection step at the time of writing.

B. Training

All learned models are trained from scratch with Adam [30] at a learning rate of 10^{-3} , cosine decay to zero without warm-up, batch size 32, 100 epochs, and mixed-precision arithmetic. Augmentation is a random resized crop and a horizontal flip. Unless stated otherwise, the channel SNR during training is fixed at 7 dB, and the checkpoint with the highest validation accuracy at 7 dB is kept. Final DJSCC models were trained for three paired training and channel seeds at each bandwidth ratio, and the task-aware digital model for three seeds at $r = 1/6$. The SNR-randomized and PAPR-capped variants were trained once each. All results in this paper come from the first seed pair, except the three-seed diagnostic in Section VII-C.

C. Digital operating-point selection

For each SNR and bandwidth ratio, the JPEG 2000 system selects an encoding resolution (a downsampled image size, upsampled bicubically at the receiver), an LDPC rate from $\{1/3, 1/2, 2/3, 5/6\}$, and a modulation from BPSK, QPSK, and 16-QAM. The JPEG 2000 codestream is the largest one that fits in the transport-block payload. Selection maximizes the expected number of correct classifications on the validation split,

$$\mathbb{E}[A] = P_s A_{\text{codec}} + (1 - P_s) A_{\text{out}}, \quad (12)$$

where P_s is the probability that every code block decodes, taken from the measured BLER curves; A_{codec} is the measured validation accuracy of the fine-tuned classifier on error-free decoded images at that setting; and A_{out} is the measured accuracy of the outage rule. Exact ties are broken in favor of the more reliable link, then the lower-order modulation, the lower code rate, and the larger image. The outage rule predicts a fixed class (class 0) whenever a packet fails or no feasible encoding exists. On the class-balanced validation split this rule scores exactly 10.0%, and every failed transmission stays in the accuracy denominator.

The selection is analytic, but the reported accuracy is not: every selected operating point is re-run image by image through the full chain, and the realized outcome is what is reported. The learned models are trained once and are not adapted to the test SNR. This asymmetry favors the digital baseline, and it is intentional.

D. Paired evaluation and statistical target

Each system is evaluated on the same 1000 validation images at 21 SNRs: -8 to 7 dB in 1 dB steps, then 9, 11, 13, 15, and 18 dB. The AWGN realization for each image depends only on the image identity, the SNR, and k , so systems at the same bandwidth ratio see the same noise wherever their representations allow it.

The primary statistical target is the paired, image-level difference in top-1 accuracy between systems at a fixed SNR and bandwidth ratio. Stable sample IDs are used as the resampling unit so that system and SNR comparisons remain paired within each bootstrap draw. The analysis procedure is configured for 10,000 paired bootstrap resamples and averages over the three seed pairs before resampling. Four hypotheses were preregistered. The primary one (H1) requires the paired DJSCC-minus-digital difference at $r = 1/6$ to be significantly positive at three or more consecutive SNRs at or below 7 dB. The others concern the size of the digital cliff, convergence of the two systems at high SNR, and whether DJSCC also exceeds the task-aware digital control. All four are decided on the test split. The validation results below do not decide any of them.

Bandwidth-ratio selection for the classical system is based on validation data rather than on learned-versus-classical test results. This keeps the choice of operating point separate from the comparison it is later used to evaluate. The ratio $1/6$ was selected as the main comparison ratio and $1/24$ as the low-bandwidth ratio.

VI. IMPLEMENTATION CHECKS

A. Reference classifier and learned-model profile

The clean ResNet-18 reaches 898/1000 on the validation split after 100 epochs, above the target of 0.88. It has 11,181,642 parameters.

At $r = 1/2$ the DJSCC model has 1,640,957 parameters. A batch-32 profile on an RTX 4060 Laptop GPU measured 1.004 GiB of peak reserved memory and 48.7 s per epoch, or about 1.35 h for 100 epochs.

B. LDPC conformance and BLER characterization

The TS 38.212-derived LDPC core was checked against an independent reference implementation with $K = 128$, $N = 256$, base graph 2, lifting size 22, rate $1/2$, offset min-sum decoding, and 5000 blocks per SNR point. Across BPSK, QPSK, and 16-QAM, the waterfall at BLER 10^{-2} is within 0.0037 dB of the reference. Packetization was checked across 216 configurations, of which 215 are feasible and one is structurally infeasible under the configured budget.

The BLER curves used for selection cover 3213 physical-layer operating points in 153 curves, at 5000 trials each (16,065,000 block trials in total). Operating points without a measured curve are excluded from selection rather than assigned a default BLER.

TABLE III
IMPLEMENTATION CHECKS.

Item	Result
Clean classifier	0.898 validation top-1 (898/1000), ResNet-18 trained from scratch
DJSCC profile	1,640,957 parameters at $r = 1/2$; 1.004 GiB peak reserved VRAM at batch 32
LDPC conformance	Waterfall offset ≤ 0.0037 dB at BLER 10^{-2} against the reference decoder
BLER campaign	153 curves; 3213 points; 5000 trials/point; 16,065,000 trials total

C. Classical validation sweep

The classical validation sweep evaluated 288 structural configurations over all 1000 validation images, for 288,000 image-configuration evaluations. Of these, 264,000 were delivered and 24,000 were codec-infeasible, with no decode or structural failures. Selection produced a valid operating point for all 378 requested SNR cells.

VII. VALIDATION RESULTS

All results in this section are counts out of the same 1000 validation images, from one training seed pair. Table IV gives the accuracy of every variant at selected SNRs. Differences of one or two percentage points between curves are within what a different seed or noise draw could change and should not be read as rankings.

A. DJSCC and the adaptive JPEG 2000 baseline

Fig. 1 shows the main comparison at $r = 1/6$. DJSCC produces a usable prediction at every SNR, with no delivery failures by construction. Its accuracy falls from 83.8% at 9 dB to 72.8% at -8 dB, a loss of 11 points across the 17 dB range.

The JPEG 2000 system delivers nothing from -8 to -5 dB and scores the outage floor of 10.0%. At -4 dB every packet decodes and accuracy jumps to 83.4%, above DJSCC's 79.4% at the same SNR. From -4 dB upward the digital system is ahead at every measured point, and the gap widens to about six points at high SNR (89.3% against 83.4% at 18 dB). The measured transition lies between -5 and -4 dB. No SNR between those two points was evaluated.

Table V lists the operating points the baseline selected. Just above the transition it sends a 64×64 image with BPSK and a rate-1/3 code, the most robust setting available, and it moves to larger images and higher-order modulation as the channel improves. Two points deliver less than all packets. At -2 dB, 23 of 1000 packets fail, and at 9 dB, 38 fail. At 9 dB the selected setting (16-QAM, rate 2/3, full resolution) delivers every packet at 11 dB, so the 9 dB point sits on the steep part of that setting's BLER curve. There, the analytic prediction and the image-by-image run disagree by a few percent of packets.

B. Bandwidth ratio

Fig. 2 repeats the comparison at $r = 1/24$, with 3200 channel uses per image. DJSCC loses most at low SNR: at -8 dB it falls from 72.8% to 49.4%, while at 18 dB it drops only from

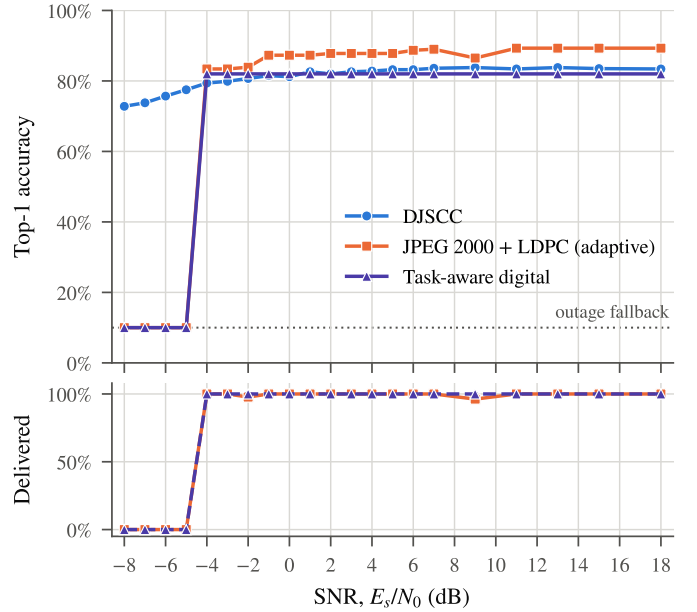


Fig. 1. Validation accuracy (top) and fraction of packets delivered (bottom) at $r = 1/6$. DJSCC is scored by its own task head, and JPEG 2000 by the ResNet-18 fine-tuned on JPEG 2000 artifacts. The dotted line is the 10% outage rule. The digital systems re-select their operating point at every SNR on validation data; DJSCC is trained once at 7 dB. Lines join measured points only.

83.4% to 82.8%. The JPEG 2000 system delivers nothing up to -3 dB. At -2 dB it delivers 82% of packets, but the images that arrive are heavily compressed and only 34.8% of them are classified correctly. DJSCC stays ahead up to +3 dB, the JPEG 2000 system is ahead from +4 to +7 dB, and DJSCC is marginally ahead again at 9 dB (82.7% against 82.4%), where 62 digital packets fail. Above 9 dB the digital system leads by 3.8 to 4.9 points. At this ratio the two curves cross more than once, and no single crossover SNR describes them.

C. Task-aware digital control

The task-aware digital system (Fig. 1) behaves like the JPEG 2000 system on the delivery axis. It fails completely up to -5 dB and delivers every packet from -4 dB. Once its packets arrive, its accuracy is 82.0% at every SNR, because error-free delivery reproduces the transmitter's quantized features exactly.

Against the JPEG 2000 system, the delivered features score 82.0% and the delivered images 83.4% to 89.3%. That difference cannot be attributed to the representation: the features are read by a single linear layer, the images by a fine-tuned ResNet-18 with about seven times as many parameters as the whole DJSCC model. Against DJSCC, in the seed pair used throughout this section the task-aware digital system is 2.6 points ahead at -4 dB, the two cross between 0 and 2 dB, and DJSCC is 0.8 to 1.8 points ahead from 4 dB upward. A precision diagnostic over all three seed pairs, with 10,000 paired bootstrap resamples, does not support the single-seed lead: averaged over the three seed pairs the two systems are tied at -4 and -3 dB (DJSCC minus digital between -0.4 and

TABLE IV

VALIDATION TOP-1 ACCURACY (%) AT SELECTED SNRS, 1000 IMAGES PER ENTRY. ENTRIES OF 10.0 AT LOW SNR ARE THE OUTAGE RULE APPLIED TO EVERY IMAGE. THE DIGITAL SYSTEMS RE-SELECT THEIR OPERATING POINT AT EVERY SNR; THE LEARNED MODELS ARE TRAINED ONCE.

r	System	SNR E_s/N_0 (dB)								
		-8	-6	-4	-2	0	4	7	9	18
1/6	DJSCC	72.8	75.7	79.4	80.7	81.2	82.8	83.6	83.8	83.4
	DJSCC, SNR-randomized	77.0	80.2	82.4	82.5	83.0	82.9	83.8	84.1	83.9
	DJSCC, PAPR-capped	75.9	78.8	80.5	80.7	82.3	82.9	82.8	83.0	83.2
	DJSCC reconstruction, clean classifier	44.7	54.5	61.1	63.9	69.0	74.1	76.1	77.5	76.8
	JPEG 2000 + LDPC, adaptive	10.0	10.0	83.4	83.9	87.3	87.8	89.0	86.5	89.3
	same outputs, clean classifier	10.0	10.0	60.9	78.8	84.8	87.0	88.6	86.3	89.2
	JPEG 2000 + LDPC, QPSK only	10.0	10.0	10.0	10.0	87.3	87.8	88.7	88.7	88.7
	JPEG 2000 + LDPC, fixed MCS	10.0	10.0	10.0	10.0	10.0	10.0	89.0	89.0	89.0
	Baseline JPEG + LDPC	10.0	10.0	67.8	76.7	84.0	87.2	87.9	10.0	88.6
	Task-aware digital	10.0	10.0	82.0	82.0	82.0	82.0	82.0	82.0	82.0
Label transmission	10.0	10.0	82.0	82.0	82.0	82.0	82.0	82.0	82.0	
1/24	DJSCC	49.4	61.7	69.3	76.2	78.7	82.0	82.9	82.7	82.8
	JPEG 2000 + LDPC, adaptive	10.0	10.0	10.0	31.0	69.4	83.4	85.8	82.4	87.7
	same outputs, clean classifier	10.0	10.0	10.0	14.2	26.2	60.9	80.6	80.7	87.0

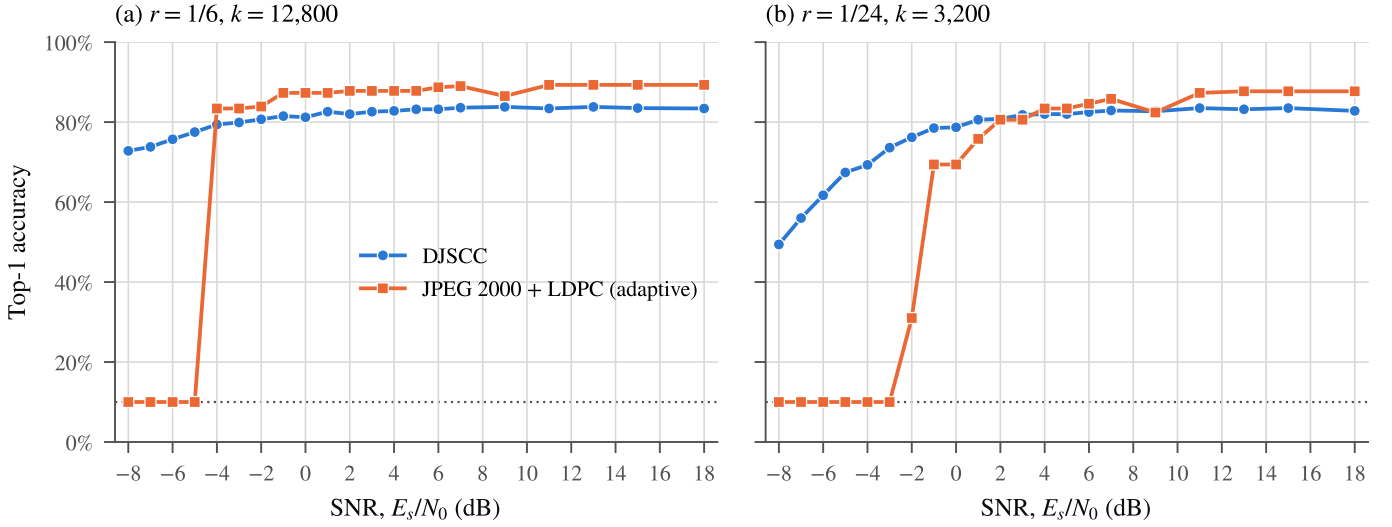


Fig. 2. DJSCC against the adaptive JPEG 2000 baseline at two bandwidth ratios. Reducing the channel uses fourfold moves the point where the digital system overtakes DJSCC from -4 dB to $+4$ dB. Scorers and selection as in Fig. 1.

TABLE V

OPERATING POINTS SELECTED BY THE ADAPTIVE JPEG 2000 BASELINE AT $r = 1/6$, WITH THE REALIZED VALIDATION ACCURACY AND DELIVERED FRACTION.

SNR (dB)	Image	LDPC	Mod.	Acc. (%)	Deliv. (%)
-8 to -5	160	1/3	BPSK	10.0	0.0
-4, -3	64	1/3	BPSK	83.4	100.0
-2	96	1/2	BPSK	83.9	97.7
-1, 0	128	1/3	QPSK	87.3	100.0
1	128	2/3	BPSK	87.3	100.0
2 to 5	128	1/2	QPSK	87.8	100.0
6	160	5/6	QPSK	88.7	100.0
7	160	1/2	16-QAM	89.0	100.0
9	160	2/3	16-QAM	86.5	96.2
11 to 18	160	2/3	16-QAM	89.3	100.0

system. At this ratio the main difference between the two task-aware systems is therefore whether a prediction reaches the receiver at all, more than the quality of the representation.

The label transmission system scores 82.0% wherever it delivers, identical to the task-aware digital system, because it sends that system's own prediction. It also fails below -4 dB. That is a property of the implementation: the one-byte label is carried in the task-aware system's 4096-bit packet, so it inherits that packet's decoding threshold. A link built for a 4-bit payload could use a much lower code rate and would be expected to survive at lower SNR. This variant therefore bounds the accuracy of transmitting a decision, not the SNR at which a decision can be transmitted.

D. Effect of the receiver classifier

Fig. 3(a) scores the same JPEG 2000 outputs with both classifiers. Above 4 dB the two agree within one point. Just

+0.1 points), and DJSCC is 0.6 to 1.8 points ahead from -2 to 7 dB. Below -4 dB it has the same cliff as the JPEG 2000

above the delivery transition they do not: at -4 dB the fine-tuned classifier scores 83.4% and the clean one 60.9%. With the clean classifier, DJSCC would lead the JPEG 2000 system up to -2 dB instead of -5 dB. At $r = 1/24$ the difference is larger, 69.4% against 26.2% at 0 dB, and with the clean classifier DJSCC would lead up to 9 dB. Near the cliff the baseline sends small, heavily compressed images, and a classifier trained on such images is worth up to 43 points, consistent with the finding that retraining on compressed images recovers most of a recognition model's loss [24]. The crossover SNR reported for this kind of comparison therefore depends on the classifier at the receiver as well as on the transmission scheme.

Fig. 3(b) holds the DJSCC transmission fixed and changes the receiver. Classifying the DJSCC reconstruction with the clean ResNet-18 scores 44.7% at -8 dB and 76.8% at 18 dB, well below the jointly trained task head at every SNR. The reconstruction is trained with $\lambda = 3$ and is blurred at low SNR. The task head reads the decoder features directly and does not depend on image quality in the same way.

E. Training variants: SNR randomization and peak power

Fig. 4 compares the three DJSCC models at $r = 1/6$. Training at randomized SNRs raises low-SNR accuracy: 77.0% against 72.8% at -8 dB, and 82.4% against 79.4% at -4 dB. At high SNR the two are within one point. The randomized model scores higher at 19 of the 21 SNRs and equal at one, but the two are separate training runs, so a single seed cannot show how much of the difference comes from the training distribution.

The fixed-SNR DJSCC encoder has a mean symbol-domain PAPR of 13.2 dB and a maximum of 20.1 dB over the validation images. For comparison, the adaptive JPEG 2000 system at 18 dB, which uses 16-QAM, has a mean of 2.6 dB and a maximum of 3.3 dB. The PAPR-capped model has a maximum of 3.000002 dB, within the 3 dB cap and its 10^{-4} dB numerical tolerance. Its accuracy is close to that of the unconstrained model: 75.9% against 72.8% at -8 dB and 83.2% against 83.4% at 18 dB, higher at 13 SNRs and lower at 6. As with the randomized model, the two are separate training runs, so this does not isolate the cost of the cap. PAPR here is measured on the complex symbols, not on an oversampled, pulse-shaped waveform, so it does not establish the peak power of a transmitted RF signal.

F. Restricted digital baselines

Fig. 5 shows what the adaptivity of the baseline is worth. Restricting the modulation to QPSK delays the first delivery from -4 to -1 dB and lowers the high-SNR plateau to 88.7%. A single fixed operating point (16-QAM, rate 1/2) delivers nothing below 7 dB and reaches 89.0% above it. This fixed curve is the classical cliff in its sharpest form: a drop of 79 points between 7 and 6 dB. A baseline with a single operating point would have made DJSCC look far better than the adaptive baseline does.

Replacing JPEG 2000 with baseline JPEG costs up to 15.6 points just above the transition (67.8% against 83.4% at -4 dB),

with smaller differences at higher SNR. This is consistent with JPEG's larger header overhead, which takes a large share of a small payload. The fine-tuned classifier was trained on JPEG 2000 artifacts, not JPEG artifacts, which also counts against the JPEG curve. At 9 dB the selected JPEG operating point failed on every image. That setting delivers every packet at 11 dB, so the 9 dB point again lies on the steep part of its BLER curve. The measurement is reported as observed.

VIII. DISCUSSION

On the validation split, the results support the qualitative picture that motivated the study, but only over a narrow range. DJSCC degrades gradually and the separated system has a cliff. At $r = 1/6$, however, the cliff lies between -5 and -4 dB, and above it the adaptive digital baseline is more accurate than DJSCC by 2.7 to 6.1 points. At this ratio DJSCC wins exactly where the digital system delivers nothing. At $r = 1/24$ DJSCC also wins from -1 to $+3$ dB, where the digital packets arrive but carry too few bytes for the classifier to do well on the decoded image.

Much of the high-SNR gap reflects a difference in accuracy ceilings. DJSCC with its jointly trained head saturates near 84%. The fine-tuned ResNet-18 on well-delivered JPEG 2000 images reaches 89%, close to the 89.8% of the clean classifier on uncompressed images. The DJSCC task head is a single linear layer on pooled decoder features. A deeper head, or a stronger backbone of the kind used in recent transformer-based JSCC work [9], might narrow this gap. We did not test that, and the parameter count of the learned model was deliberately kept well below that of the reference classifier.

The task-aware digital control changes the attribution question. It reaches 82.0% whenever its packets decode, so a digital system that sends learned features performs close to DJSCC above the cliff. Below the cliff it fails in the same way as the JPEG 2000 system. At this operating point, most of what DJSCC gains at low SNR comes from avoiding outage, not from a better task representation. The H4 test on the sealed test split will decide whether any residual advantage of joint coding over the task-aware digital system is distinguishable from zero. The three-seed validation diagnostic puts the smallest difference detectable with 80% power at 1.8 to 1.9 points from -4 to 7 dB and about 4 points below the cliff. That figure is pointwise and does not certify the power of the full H4 procedure, so a null result there would not show that joint coding has no advantage of a point or two.

The comparison also shows how sensitive such conclusions are to the baseline. Holding the transmission fixed, the choice of receiver classifier moves the crossover by 3 dB at $r = 1/6$. Removing adaptivity moves it by 11 dB. Replacing the codec with JPEG lowers the digital curve by up to 15 points. Each of these choices would have made DJSCC look better. We made the baseline as strong as the preregistered search space allowed. At $r = 1/6$ the learned system then wins only where the baseline cannot deliver; at $r = 1/24$ it also wins where the baseline delivers too few bytes for the classifier to do well.

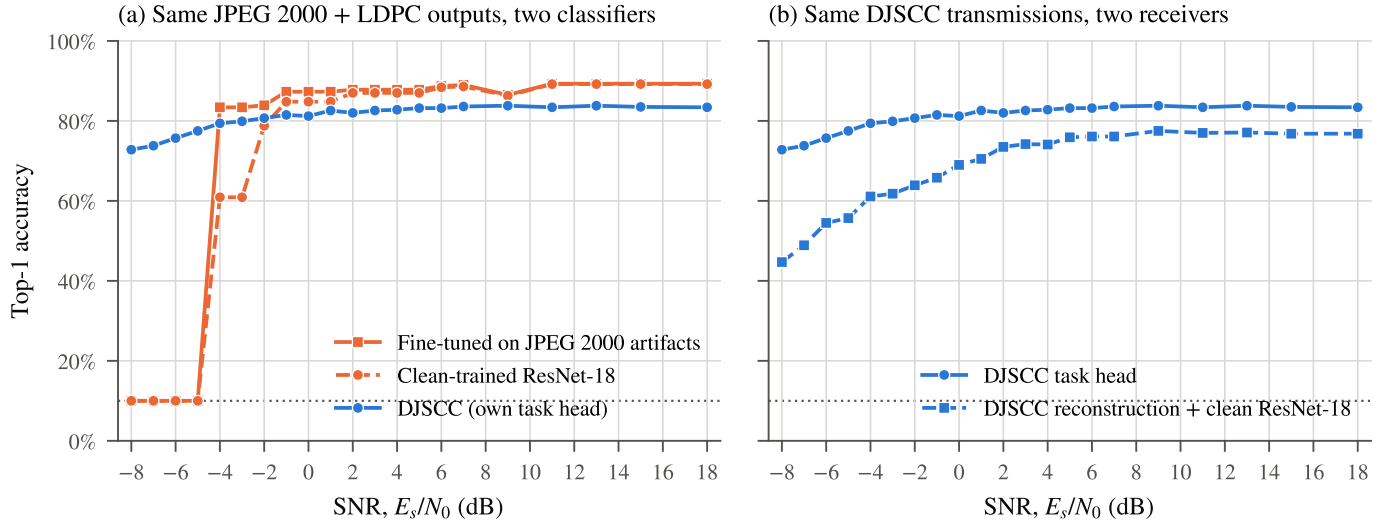


Fig. 3. (a) The same JPEG 2000 + LDPC outputs at $r = 1/6$ scored by the fine-tuned and the clean ResNet-18, with DJSCC for reference. (b) The same DJSCC transmissions read by the DJSCC task head and by the clean ResNet-18 applied to the DJSCC reconstruction.

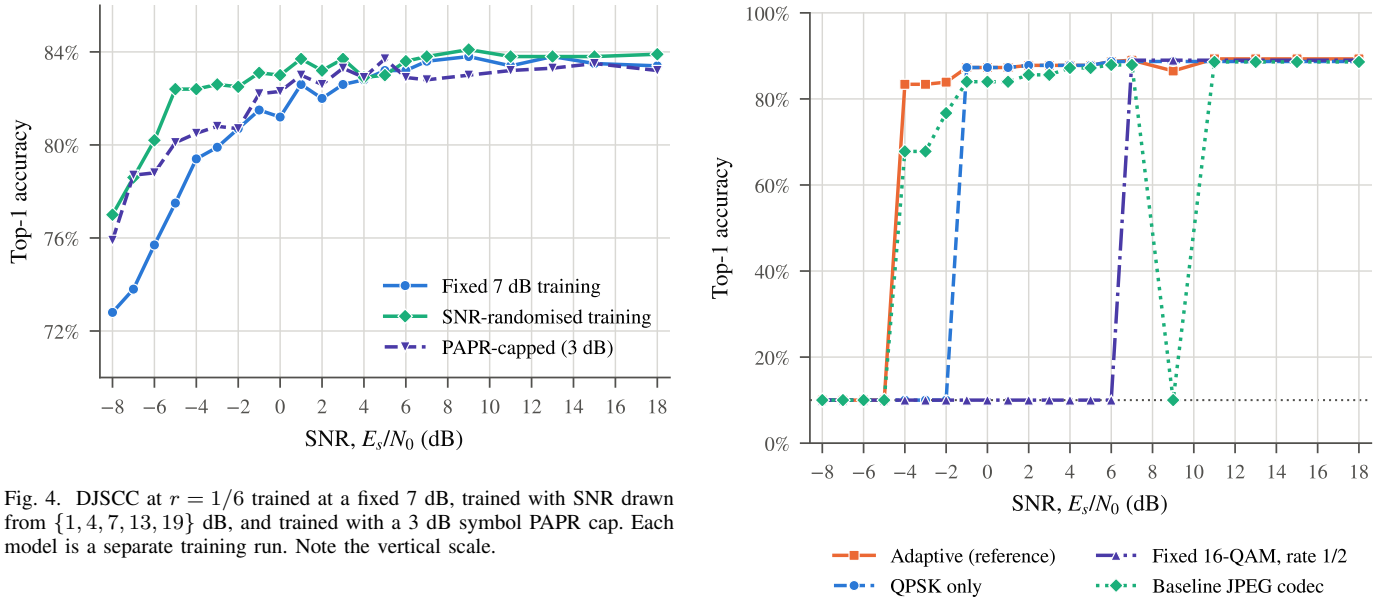


Fig. 4. DJSCC at $r = 1/6$ trained at a fixed 7 dB, trained with SNR drawn from $\{1, 4, 7, 13, 19\}$ dB, and trained with a 3 dB symbol PAPR cap. Each model is a separate training run. Note the vertical scale.

IX. LIMITATIONS

All results in this paper are from the validation split. The same 1000 images were used to choose the digital operating points, the DJSCC checkpoint, the task-aware feature dimension, and λ , so every curve is somewhat optimistic, and the digital curves more so, because they are selected per SNR. The final numbers will come from the 3925-image test split, which has not yet been used.

Apart from the three-seed diagnostic in Section VII-C, the results come from one training seed and one channel seed, and no confidence intervals are given here. The paired bootstrap described in Section V needs per-image outcomes over three seed pairs and will be run on the test split. Point differences of one or two points should not be interpreted.

The learned and digital systems are scored by different

Fig. 5. Adaptive JPEG 2000 + LDPC at $r = 1/6$ against restricted versions: QPSK only (rate and resolution still adapted), one fixed operating point at all SNRs, and baseline JPEG in place of JPEG 2000. All scored by the fine-tuned ResNet-18.

classifiers. The headline comparison is therefore a whole-system comparison and not a comparison of channel codes. Section VII-D reports both scorers for the digital outputs so that the effect can be seen.

The channel is simulated complex AWGN with perfect synchronization and known SNR at the receiver. Fading, carrier and timing offsets, and hardware nonlinearity are not modeled, and PAPR is measured in the symbol domain only. The systems are compared at equal channel uses and equal average symbol energy; this is a statement about accuracy at equal resources, not

a measured energy saving. Computational cost at the transmitter and receiver is not compared.

X. CONCLUSION

We built a controlled comparison of three ways to send an image to a remote classifier: JPEG 2000 over a 5G NR LDPC link with per-SNR adaptation, learned task features over the same link, and task-aware DJSCC, all with the same number of complex channel uses. On Imagenette-160 validation data at $r = 1/6$, DJSCC is the only system that works at or below -5 dB, where it keeps 73% to 78% accuracy. From -4 dB upward the adaptive digital baseline is more accurate by 2.7 to 6.1 points. At $r = 1/24$ the DJSCC advantage extends to $+3$ dB. At $r = 1/6$ the task-aware digital control is within two points of DJSCC across three seeds wherever it delivers, which indicates that avoiding the outage cliff accounts for most of DJSCC's low-SNR gain.

Three items remain. The preregistered test evaluation will score all systems on the sealed 3925-image test split over three seed pairs, with paired bootstrap intervals, and will decide hypotheses H1 to H4. The label-transmission bound needs a link sized for its payload before it can say how far a transmitted decision reaches. Hardware experiments with a software-defined radio, which would measure real waveforms and impairments, are planned as optional extensions of this work.

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